PROJECT REVIEW 1

SmartQMS: Intelligent Workflow-Aware Hospital Queue Management and Dynamic Resource Allocation System

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ABSTRACT

Modern healthcare facilities encounter severe operational bottlenecks in Outpatient Departments (OPD), diagnostic laboratories, and pharmacy counters due to escalating patient volumes and disjointed administrative workflows. Traditional hospital queuing systems predominantly employ static First-Come-First-Served (FCFS) or isolated single-department token ticketing. Such paradigms fail to account for clinical urgency, inter-departmental patient trajectories (e.g., Physician Consultation → Diagnostic Laboratory → Radiology → Pharmacy), varying counter service rates, and unpredictable patient no-shows. Consequently, patients endure prolonged and fragmented waiting times, clinical triage protocols are compromised, counter resources experience alternating cycles of congestion and idle time, and hospital staff face substantial cognitive overhead.

Existing research has addressed isolated aspects of healthcare resource management, including statistical arrival forecasting, static appointment scheduling, discrete-event emergency room simulation, and basic digital kiosk ticketing. However, conventional systems treat each departmental queue in complete isolation and lack real-time awareness of multi-stage clinical workflows, cross-departmental transfer latency, and dynamic urgency re-prioritization.

To address these critical limitations, this project presents **SmartQMS**, an end-to-end, workflow-aware, and priority-driven intelligent hospital queue management system. SmartQMS integrates: (1) contactless QR-based self-service check-in for appointments and walk-in arrivals; (2) dynamic triage-aware priority scheduling that mathematically balances clinical severity, scheduled appointment windows, cumulative dwell time, and emergency overrides; (3) multi-stage inter-departmental workflow orchestration that tracks and routes patients through multi-tier diagnostic and consultation pathways without requiring manual re-ticketing; (4) adaptive wait-time forecasting calibrated by active counter capacity, historical service velocity, and confidence bounds; (5) real-time event-driven synchronization utilizing WebSockets for live public hall displays, counter interfaces, and instant patient notifications; and (6) context-aware conversational AI guidance powered by Large Language Models (LLMs) to answer queue status, department directions, and preparation instructions.

Experimental evaluation plans benchmark SmartQMS against standard FCFS, static appointment booking, and uncoordinated departmental queues across key operational metrics: Average Patient Dwell Time, Counter Utilization Efficiency, Triage Latency, Inter-Department Transfer Delay, and Patient Satisfaction. The proposed approach aims to diminish patient wait times, eliminate counter starvation, and establish an explainable, data-driven healthcare flow.

KEYWORDS

Hospital Queue Management, Healthcare Operations, Priority Triage Scheduling, Multi-Department Workflow Orchestration, QR Tokenization, Real-Time WebSockets, Adaptive Wait-Time Estimation, Outpatient Department (OPD), Clinical Flow Optimization, Smart Hospital.

1. INTRODUCTION

Healthcare operations worldwide face unprecedented challenges in managing patient flow across complex hospital ecosystems. The outpatient journey is inherently dynamic, heterogeneous, and non-linear. A single outpatient visit rarely involves a single queue; instead, patients transition across multiple interconnected service points—including reception desks, general consultation chambers, specialized diagnostic laboratories (hematology, biochemistry), radiological suites (X-ray, ultrasound, CT scans), and outpatient dispensaries.

In conventional healthcare facilities, resource allocation relies either on physical queuing or rudimentary electronic token dispensers that operate on a rigid First-Come-First-Served (FCFS) basis. When hospital traffic surges during morning peak hours, several failure modes manifest:

- **Cascading Inter-Departmental Bottlenecks:** Delays in initial consultations propagate downstream, causing sudden bursts of arrivals at laboratory sample collection counters and imaging suites.
- **Triage Inefficiencies:** A clinically unstable patient or an elderly patient with acute distress is forced into the same linear queue as a routine health check-up walk-in.
- **Doctor & Counter Idling vs. Patient Crowding:** While certain specialist rooms face overflowing waiting rooms, adjacent diagnostic booths remain underutilized due to lack of synchronized scheduling.
- **High Cognitive Stress and Anxiety:** Lack of real-time visibility into queue positions, estimated consultation times, and subsequent procedural steps causes high patient agitation and frequent non-adherence or no-shows.

A concrete illustration clarifies the operational gap: Suppose a patient arrives with suspected cardiovascular distress. Under traditional setups, the patient takes a physical token for General Medicine, waits 45 minutes, is evaluated, and is referred for an urgent 12-lead ECG and Troponin blood panel. The patient must walk to the Central Laboratory, pull another physical token, and wait in an uncoordinated 30-person queue alongside routine blood sugar test patients. After blood draws, the patient walks to the Radiology wing for an echocardiogram, repeating the queue process from scratch. By the time diagnostic results arrive, the initial doctor's shift may have concluded, or the consultation queue has backed up by dozens of patients. The fundamental gap is that traditional systems treat each departmental queue as an independent, isolated island, ignoring the continuous clinical trajectory of the patient and failing to dynamically elevate priority based on clinical urgency and elapsed dwell time.

The main objectives of this research are:

1. To study and model multi-stage patient pathways across diverse departments (Consultation, Lab, Radiology, Pharmacy).
2. To formulate and implement a dynamic priority triage algorithm that combines clinical acuity, appointment punctuality, age-based vulnerability, and cumulative wait time.

3. To design a contactless cryptographic QR tokenization mechanism for instant kiosk check-in, tracking, and handoffs.
4. To calculate an adaptive, rolling wait-time estimate with confidence bounds calibrated by active counter capacity and historical service velocity.
5. To engineer a real-time bidirectional synchronization bus using WebSockets to broadcast live queue updates across public TV displays, doctor chambers, and receptionist consoles.
6. To embed a context-grounded LLM conversational agent to provide personalized queue guidance, department navigation, and preparation instructions.

Beyond the primary objectives, this study also examines several secondary research questions: whether dynamic, multi-factor priority scoring reduces clinical emergency response latency without starving standard appointment patients; how much inter-departmental transfer latency is eliminated when downstream departments are pre-notified of incoming patient workflows; what trade-off exists between strict appointment time slots versus dynamic walk-in queue interleaving under volatile patient arrival rates; and whether rolling historical service time analytics provide reliable wait-time estimates under non-deterministic consultation durations.

2. LITERATURE SURVEY / RELATED WORKS

Fifteen recent peer-reviewed works (2022–2026) covering healthcare queuing, outpatient flow scheduling, priority triage algorithms, multi-department clinical workflows, and smart hospital architectures were examined to set the stage for this research.

2.1 Multi-Priority Queuing Models in Emergency Departments

Al-Refaie et al. (2023) formulated a multi-priority queuing model to minimize patient waiting times in emergency departments based on Emergency Severity Index (ESI) classifications. The study proved that non-preemptive priority queuing drastically shortens critical patient waiting. However, their model was strictly confined to single-stage emergency rooms and did not model multi-stage outpatient workflows spanning multiple clinical departments.

2.2 Discrete-Event Simulation for Outpatient Flow Optimization

Zhang and Wang (2024) implemented an Arena-based discrete-event simulation model to analyze bottleneck stages in a tertiary hospital's outpatient department. The work demonstrated that reallocating consultation rooms during peak windows reduced overall waiting time by 22%. Nevertheless, the simulation operated offline on static historical distributions and lacked real-time feedback or live token tracking mechanisms.

2.3 IoT-Enabled Smart Hospital Token Systems

Suresh and Sundararajan (2023) designed an RFID- and IoT-based digital token management system for hospital OPDs. The system successfully replaced paper slips with digital displays. However, token progression was governed strictly by static FIFO order without considering appointment urgency, doctor availability, or multi-departmental patient handoffs.

2.4 Dynamic Outpatient Appointment Scheduling Under Unpunctuality

Liu et al. (2025) proposed a stochastic optimization framework to handle patient unpunctuality and no-shows in clinic appointments using dynamic buffer intervals. While effective at reducing doctor idle

time within a single clinic, the method assumed fixed, isolated appointments and did not address walk-in patients or cross-department diagnostic routing.

2.5 Multi-Stage Clinical Workflow Orchestration

Kavitha and Mohan (2024) investigated workflow pipeline coordination between oncology outpatient clinics and central diagnostic laboratories using Petri nets. They highlighted the critical synchronization barrier between test ordering and result review. However, their system remained conceptual and lacked interactive digital kiosks, real-time push notifications, and counter-level operator controls.

2.6 Machine Learning for Patient Wait-Time Prediction

El-Banna et al. (2023) utilized Random Forest and XGBoost regressors to predict outpatient waiting times in public hospitals based on weather, arrival hour, and historical doctor throughput. While achieving a low Mean Absolute Error (MAE), their solution was purely observational and did not integrate directly with queue dispatchers to actively reorder tokens or balance counter loads.

2.7 Reinforcement Learning for Dynamic Clinical Resource Dispatching

Chen et al. (2025) deployed a Deep Q-Network (DQN) agent to dynamically allocate examination rooms to incoming patients in a diagnostic imaging center. The agent learned effective load-balancing policies. However, its high training complexity and opaque black-box decisions make it difficult to justify to clinical staff compared to transparent, explainable priority rules.

2.8 Triage-Aware Queue Scheduling in Polyclinics

Rahman and Hasan (2024) developed an automated triage classification engine that mapped patient vital signs to priority levels using rule-based heuristics. The work confirmed that automated triage reduced adverse clinical events in outpatient waiting lobbies. Yet, it treated priority as static once assigned, causing prolonged starvation for low-priority patients during peak influx.

2.9 Real-Time Event-Driven Hospital Telemetry Dashboards

Nair et al. (2026) constructed a microservices-based hospital telemetry system using Apache Kafka and WebSockets to monitor ward occupancy and patient transit. The architecture demonstrated the feasibility of sub-second state dissemination across large medical campuses. However, it functioned solely as a passive monitoring layer without active queue governance or token dispatch algorithms.

2.10 Hybrid Walk-in and Appointment Interleaving Framework

Gupta and Mehrotra (2022) proposed a mathematical interleaving algorithm that reserved dedicated slots for pre-booked appointments while dynamically absorbing walk-in patients. The model proved mathematically optimal for single-server queues but struggled when scaled to multi-server, multi-department networks where appointments depend on prior test completions.

2.11 Patient No-Show Mitigation via Automated Push Reminders

Fernandez et al. (2023) conducted an empirical study demonstrating that automated SMS and app notifications reduced clinic no-show rates by 31.4%. However, the notification trigger was strictly time-scheduled (e.g., 24 hours prior) rather than reactive to live queue velocity and real-time waiting room crowding.

2.12 Inter-Departmental Patient Handoff Protocols

Patel and Joshi (2025) modeled patient handoffs between emergency departments, intensive care, and surgical wards using queuing network theory. The authors highlighted that uncoordinated inter-department transfers account for over 40% of preventable clinical delays. However, their focus was inpatient transfer rather than ambulatory outpatient multi-department navigation.

2.13 Comprehensive Survey on Healthcare Queuing Systems

Lakshmi and Iyer (2023) presented an extensive taxonomy categorizing healthcare queuing literature across analytical queuing models, simulation, meta-heuristics, and machine learning. The survey emphasized that fewer than 8% of published architectures bridge the gap between algorithmic theory and production-ready, full-stack hospital deployments.

2.14 Attention-Based Outpatient Congestion Forecasting

Bhat et al. (2024) designed a Temporal Convolutional Network (TCN) with self-attention to forecast hourly congestion across hospital outpatient lobbies. The model accurately captured seasonal trends. However, it operated at macro-level census forecasting rather than tracking individual patient tokens and specific room assignments.

2.15 Conversational AI for Healthcare Navigation and Triage

Martinez et al. (2026) investigated Large Language Model (LLM) agents for patient orientation and preliminary symptom categorization in digital hospital portals. The study demonstrated high patient usability. However, the LLM had no access to live backend queue databases or counter statuses, leading to hallucinations regarding current waiting times and queue positions.

3. LITERATURE SURVEY TABLE

Year	Technique	Merits	Demerits	Limitation
2023	Multi-Priority Queuing (ESI Model)	Drastically reduces critical patient wait times	High starvation risk for non-urgent patients	Confined to single-stage emergency rooms; no outpatient workflow
2024	Discrete-Event Simulation (Arena)	Accurate bottleneck detection and room reallocation	Offline batch execution only	No live queue tracking or real-time token interaction
2023	IoT & RFID Smart Token System	Replaces physical paper slips; eliminates manual chaos	Static FIFO dispatch only	Lacks clinical priority scoring and multi-department routing
2025	Stochastic Appointment Scheduling	Mitigates unpunctuality and doctor idle time	Designed for isolated single clinics	Does not support walk-ins or multi-department diagnostic flows

Year	Technique	Merits	Demerits	Limitation
2024	Petri Net Workflow Orchestration	Rigorous formal modeling of clinic-lab synchronization	Lacks software implementation	No digital kiosk integration or patient-facing interfaces
2023	Random Forest / XGBoost Wait Estimation	Accurate predictive wait times using historical data	Purely passive predictive model	Not connected to active queue dispatching or token reordering
2025	Deep Q-Network (DQN) Room Dispatching	Self-learning optimal counter allocation policies	High computational cost; black-box opacity	Clinicians cannot audit or explain sudden queue order shifts
2024	Automated Polyclinic Triage Heuristics	Rapid vital-sign-based prioritization	Static priority assignment	Causes starvation for routine patients during continuous surges
2026	Kafka & WebSocket Hospital Telemetry	Sub-second real-time event dissemination	High architectural overhead	Purely observational dashboard; no queue scheduling logic
2022	Hybrid Appointment/Walk-in Interleaving	Optimal single-server queue interleaving	Degrades in multi-server environments	Cannot handle inter-departmental dependent appointments
2023	Automated Multichannel Push Notifications	Reduces patient no-shows by 31.4%	Fixed time-delay triggers	Not reactive to live queue velocity or actual room status
2025	Inpatient Queuing Network Analysis	Quantifies inter-department transfer delay costs	Focused exclusively on inpatient admissions	Fails to address outpatient ambulatory multi-stage loops
2023	Healthcare Queuing Taxonomy Survey	Comprehensive classification of algorithms	Review paper only	Proposes no new algorithmic or architectural implementation
2024	Attention-Based Congestion Forecasting (TCN)	Captures macro-level lobby congestion patterns	Coarse aggregate time-granularity	Cannot track individual tokens or specific room allocations
2026	LLM-Based Hospital Navigation Agent	High usability in guiding patient queries	No live database connectivity	Susceptible to hallucinating queue status and wait times

4. RESEARCH GAP

The reviewed literature can be grouped into five primary directions: single-stage queue simulation, static electronic token systems, passive wait-time forecasting, black-box reinforcement learning dispatchers, and disconnected information silos. Although each aspect is well studied, existing hospital queue solutions treat each clinical unit as an independent entity, ignoring the continuous multi-stage journey of the patient.

The central research gap is: How can a unified healthcare queue management platform dynamically orchestrate multi-department patient workflows, adaptively balance clinical urgency with waiting-time fairness, provide confidence-bounded wait estimates, and maintain sub-second multi-client synchronization across physical hospital counters?

The SmartQMS project focuses on combining the following key pillars:

- **Unified Multi-Stage Workflow Tracking:** Orchestrating a patient's entire visit (Consultation → Lab → Radiology → Pharmacy) under a single master workflow with automated chained queue handoffs.
- **Dynamic Fair-Priority Scoring (DFPS):** Elevating priority dynamically with elapsed dwell time to eliminate patient starvation while honoring clinical triage.
- **Contactless Self-Service Tokenization:** Cryptographic QR code generation and instant scanning across touch kiosks.
- **Adaptive Rolling Wait-Time Estimation:** Computing real-time wait times with confidence bounds based on live counter availability and service rates.
- **Deterministic, Explainable & Auditable Architecture:** Providing clinical staff with full control, emergency overrides, and complete event-sourced audit logs.
- **Full-Stack Production-Ready Implementation:** Delivering responsive web applications for kiosks, public displays, and staff consoles with live WebSocket synchronization and LLM conversational guidance.

5. PROPOSED SYSTEM

The proposed system introduces an intelligent workflow and priority orchestration layer between patient arrival and hospital counter service delivery. It does not replace clinical decision-making; instead, it coordinates patient routing and queue sequencing across multiple physical counters and departments.

```
Patient Arrival → QR/Kiosk Check-In → Dynamic Priority Scoring → Workflow Generation → Departmental Queue Allocation → Adaptive Wait Estimation → Real-Time WebSocket Dispatch → Service Execution → Auto-Transfer / Completion → Telemetry & Feedback
```

Step-by-Step System Flow:

1. **Patient Arrival:** Patients arrive either with pre-booked online appointments or as walk-ins at the hospital kiosk.
2. **QR Check-In:** The patient scans their cryptographic QR code or enters their phone number at the self-service kiosk to instantly generate a departmental token.

3. **Dynamic Priority Scoring:** The engine calculates the patient's initial priority score based on triage acuity, appointment punctuality, age vulnerability, and initial wait time.
4. **Multi-Stage Workflow Generation:** For multi-department visits, a master `PatientWorkflow` is instantiated with chained `WorkflowStep` records.
5. **Queue Allocation & Wait Estimation:** The patient is placed into the active queue of the starting department. The system calculates the estimated waiting time and confidence interval.
6. **Live Dispatch & Notification:** When a counter operator invokes `callNext`, WebSockets immediately update public hall TV monitors with chime audio and designate the room number.
7. **Service Execution:** During consultation, the room status is locked to `OCCUPIED`. Clinicians can prescribe diagnostic tests, directly appending steps to the workflow.
8. **Automated Transfer & Completion:** When service completes, the system frees the room and automatically transfers the patient to the next scheduled department queue (e.g., Lab or Radiology) without requiring manual re-registration.

6. SYSTEM ARCHITECTURE

The architecture places a lightweight, reactive decision layer between client interfaces and the persistent clinical database, maintaining sub-100ms real-time synchronization. The overall component flow is:

```
Client Tier (Kiosks, Displays, Staff Portals) ↔ API Gateway / Express Controller
↔ Queue & Priority Engine ↔ Workflow Orchestrator ↔ Socket.IO Real-Time Engine ↔
Prisma ORM ↔ PostgreSQL Database
```

The key architectural components are:

- **Self-Service Kiosk Client:** Touch-screen web application facilitating rapid patient check-in, walk-in registration, and QR ticket issuance.
- **Public Waiting Hall Display:** High-definition TV display client rendering active called tokens, designated room numbers, audio bells, and upcoming queue positions.
- **Staff & Doctor Chamber Portal:** Role-based interface enabling doctors, nurses, lab technicians, and pharmacists to call tokens, record consultation status, and initiate transfers.
- **Queue & Dispatch Engine:** Core algorithmic service executing dynamic priority ranking, token generation, and wait-time estimations.
- **Multi-Stage Workflow Orchestrator:** Manages directed acyclic graph (DAG) transitions for patient pathways across hospital departments.
- **Real-Time WebSocket Bus:** Socket.IO server broadcasting live state changes (`token:called`, `queue:updated`, `patient:checked-in`) across dedicated departmental rooms.
- **Context-Grounded Conversational AI:** Google Gemini API integration injecting live queue context to answer patient queries regarding waiting times and hospital navigation.
- **Persistence Tier:** PostgreSQL relational database managed by Prisma ORM ensuring strict schema validation, transactions, and event logging.

7. MODULES OF THE PROPOSED SYSTEM

7.1 Patient Registration & QR Token Generation

Facilitates dual-mode entry: pre-booked online appointments and walk-in arrivals. Generates unique, tamper-evident cryptographic QR tokens (e.g., `P-2026-000124` and `C-001`, `L-001`) linked to patient profiles. Kiosks allow rapid token generation by choosing primary symptoms or departments.

7.2 Dynamic Triage & Priority Scoring Engine

Replaces rigid FIFO queues with an intelligent, dynamic priority algorithm. To prevent starvation of low-priority walk-in patients while ensuring rapid handling of critical cases, the priority score $P_i(t)$ of patient i at current time t is computed as:

$$P_i(t) = w_1 \cdot Acuity(i) + w_2 \cdot ApptPunctuality(i, t) + w_3 \cdot ln(1 + t - t_{checkin}) + w_4 \cdot AgeVulnerability(i)$$

Where $Acuity(i)$ is the clinical triage level (0 = Routine, 1 = Urgent, 2 = Emergency); $ApptPunctuality(i, t)$ rewards patients arriving within their scheduled window; $ln(1 + t - t_{checkin})$ is a time-dependent logarithmic aging factor guaranteeing that waiting time steadily elevates priority to eliminate starvation; and $AgeVulnerability(i)$ provides an adjustment for pediatric and geriatric patients.

7.3 Multi-Department Workflow Orchestrator

Models patient clinical pathways as multi-step workflows (`PatientWorkflow` → `WorkflowStep[]`). Upon consultation completion in General Medicine, diagnostic orders automatically transition the patient into the target department's queue (`LAB` or `XRAY`) under a `WAITING` status, preserving their overall hospital tenure.

7.4 Historical Queue & Telemetry Database

Persists granular state transition logs for every queue entry, tracking timestamps for check-in, call-out, service start, and service completion.

Field	Type	Description
`id`	String (UUID)	Unique identifier for the queue entry
`tokenNumber`	String	Human-readable display token (e.g., C-001, L-014)
`queueId`	String (FK)	Foreign key to the specific department queue
`appointmentId`	String (FK, Opt)	Associated scheduled appointment record
`patientWorkflowId`	String (FK, Opt)	Master multi-department workflow tracker
`workflowStepId`	String (FK, Opt)	Active clinical step in the workflow
`roomId`	String (FK, Opt)	Assigned consultation room or test booth
`priority`	Integer	Calculated dynamic priority score (0=Normal, 1=Appt, 2=Urgent)

Field	Type	Description
`status`	Enum	State: WAITING, CALLED, IN_SERVICE, COMPLETED, SKIPPED
`checkinTime`	DateTime	Timestamp when token was generated
`calledAt`	DateTime (Opt)	Timestamp when patient was called to counter
`serviceStartAt`	DateTime (Opt)	Timestamp when consultation/test started
`serviceEndAt`	DateTime (Opt)	Timestamp when service was completed

7.5 Adaptive Wait-Time Estimation Module

Calculates the expected wait time W_e for a patient with N_{ahead} patients ahead in the queue:

$$W_e = (N_{ahead} \cdot \mu_{service}) / C_{active}$$

$$\textit{Confidence Interval} = [\, W_e \cdot (1 - \alpha),\quad W_e \cdot (1 + \beta)\,]$$

Where $\mu_{service}$ is the rolling exponentially weighted moving average (EWMA) of service duration over the preceding 2 hours, C_{active} is the count of active counters, and α, β represent confidence bounds.

7.6 Doctor & Counter Allocation Engine

Provides doctors and lab technicians with an interface to call next patients, start consultations, complete services, and handle patient absences (`skipEntry`, `recallEntry`) with room state tracking (`AVAILABLE`, `OCCUPIED`, `MAINTENANCE`).

7.7 Inter-Department Transfer & Handoff

Allows clinicians to seamlessly refer patients to subsequent departments (`transferEntry`). The system completes the source entry, archives the old token, issues a new department token, and notifies the target desk.

7.8 Real-Time State Synchronization Engine

Socket.IO event bus managing bidirectional communication: emitting `queue:updated`, `token:called`, and `patient:checked-in` events to instantly sync all client dashboards and waiting hall TVs.

7.9 Public Waiting Hall Display & Kiosk

High-contrast, audio-enabled TV display showing currently serving tokens, room numbers, recent calls, and upcoming queue tokens for clear patient visibility.

7.10 Analytics, AI Assistant & Operational Dashboard

Provides hospital administrators with real-time operational metrics: peak arrival hours, wait-time distributions, counter utilization rates, and a context-aware Google Gemini AI assistant for patient queue navigation.

8. EVALUATION PLAN

The proposed method will be compared against four baselines:

1. **Random / Unregulated Walk-in:** Physical unmanaged queuing without centralized tokens.
2. **Standard FCFS Token System:** Electronic token ticketing without priority differentiation.
3. **Static Appointment Scheduling:** Fixed appointment slots without dynamic walk-in interleaving.
4. **Isolated Departmental Queues:** Separate, uncoordinated queues requiring manual re-ticketing at each stage.

Evaluation will use the following metrics:

- **Average Patient Wait Time (AWT):** Mean duration from token check-in to service start across all departments.
- **Total Hospital Dwell Time (THDT):** Elapsed time from initial hospital entry to final medication or report collection.
- **Counter Idle Ratio (CIR):** Percentage of operational hours doctor chambers remain idle while patients wait in lobbies.
- **Triage Response Latency (TRL):** Time taken from arrival to consultation for high-acuity/emergency patients.
- **Starvation Index:** Maximum wait time endured by routine walk-in patients during peak load hours.
- **Inter-Department Transfer Latency:** Time elapsed between completing one department's service and checking into the next.

9. EXPECTED OUTCOME

The proposed system is expected to significantly reduce outpatient waiting times, eliminate counter starvation, streamline multi-stage diagnostic transfers, and enhance patient satisfaction compared to conventional approaches.

Operational Dimension	Conventional Hospital Queue	SmartQMS System
Patient Registration	Manual line at reception desk (15–30 min delay)	5-second contactless QR / Self-Kiosk scan
Queue Sequencing	Blind FIFO; emergency cases delayed	Multi-factor Dynamic Priority with Aging
Multi-Stage Transitions	Patient re-queues physically at each department	Automated background transfer & pre-notification
Wait-Time Transparency	None; patients wait with high anxiety	Real-time confidence-bounded wait estimates
Counter Utilization	Unbalanced; some rooms idle while others jam	Dynamic counter load balancing and active routing
Staff Overhead	High receptionist & nurse coordination friction	Autonomous orchestration and audit trails

Research Hypotheses & Anticipated Improvements:

- 30% to 45% reduction in total outpatient hospital dwell time for multi-stage workflows.
- Over 60% reduction in emergency and high-acuity triage response latency.
- Elimination of patient starvation via logarithmic time-aging priority factors.
- Sub-100ms real-time propagation of token calls to waiting hall displays.
- Significant decrease in patient inquiries at nurse stations through contextual LLM guidance.

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